

Mihajlo Milojević

Software Engineering Student

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Summary

Passionate and driven software engineering student with hands-on experience in full-stack development, freelance projects, and programming competitions. Actively exploring different areas of software development through internships, hackathons, and team-based projects. Eager to contribute, learn, and grow in dynamic engineering environments.

Education

Faculty of Technical Sciences, University of Novi Sad

BSc in Software Engineering and Information Technologies

2023 – Present

Electro-Transport Technical School “Nikola Tesla”, Kraljevo

Electrotechnician of Information Technology

2019 – 2023

- GPA: 5.0 (maximum) – Valedictorian

Experience

Cryptography Intern

MVP Workshop – Team 3327

February – March 2025

- Researched and implemented a key recovery mechanism for the ECPDKSAP (Curvy) stealth address protocol
- Contributed to the design of distributed key generation and threshold signature schemes using MPC
- Co-developed a Go-based cryptographic library for share generation, reconstruction, and validation using ECDSA and Merkle trees; compiled to Web Assembly

React Native Developer (Part-Time)

Fitnesses

September 2022 – December 2023

- Implemented redesigned UI components and layouts in an existing cross-platform fitness app
- Developed new features based on product specifications
- Maintained and refactored legacy code for improved performance

Freelance Fullstack Developer

Remote

2022 – Present

- Built and deployed **ChillTube**, an anime streaming platform using Next.js and MySQL
- Developed **The Reptile House**, an e-commerce web app with product listings, auth, and admin tools
- Built most of the REST API for the **Belgrand** mobile app using Hono (JavaScript)
- Migrated backend services from Hono (JavaScript) to Fiber (Go) for the **Arhivix** project

Technical Skills

Languages: JavaScript, C, C++, C#, Java, Python, PHP, Go, SQL (PL/SQL)

Frameworks & Libraries: React, React Native, Next.js, Express.js, Fiber, Hono

Tools & Platforms: Git, GitHub, VS Code, Postman

Other: Cryptography, Computer Networks, REST APIs

Selected Projects

EU Quiz Mobile App

Team Project – 1st Prize Winner

GitHub: github.com/MihajloMilojevic/evropski-dnevnik

Technologies: React Native, Node.js, MongoDB

- Built a full-stack mobile application to educate users about the European Union through interactive quizzes
- Developed a RESTful API in Node.js to handle user sessions, scores, and question data using MongoDB
- Collaborated with a team of 3 to complete the project under a competition deadline
- Won first place in a national civic-tech competition

MHotelify – Hotel Management System

Java Swing Desktop Application

GitHub: github.com/MihajloMilojevic/MHotelify_OOP1

Technologies: Java, Swing, JUnit, XChart

- Created a desktop application to manage hotel operations, including room bookings and customer information
- Applied object-oriented programming principles for maintainability and scalability
- Utilized JUnit for testing and XChart for visual reporting
- Designed a user-friendly interface using Swing

Checkers.py – AI-Powered Checkers Game

Python Desktop Application

GitHub: github.com/MihajloMilojevic/Checkers.py

Technologies: Python, PyGame, Minimax Algorithm, Alpha-Beta Pruning

- Developed a checkers game with single-player (AI) and two-player modes
- Implemented an AI opponent using the Minimax algorithm with alpha-beta pruning
- Designed a GUI that highlights moves and enforces rules
- Packaged for easy Windows installation

mysql-like-mongo – Node.js ORM for MySQL

NPM Package

GitHub: github.com/MihajloMilojevic/mysql-like-mongo

Technologies: Node.js, MySQL

- Developed a Mongoose-like Node.js ORM for MySQL databases
- Enabled schema definitions with built-in CRUD and custom functions
- Published the package to NPM for developer use

The Reptile House – E-commerce Web Application

Client Project

GitHub: github.com/MihajloMilojevic/The-Reptile-House

Technologies: Next.js, React.js, Node.js, MySQL

- Developed an e-commerce platform with product browsing, authentication, and admin panel
- Built a custom REST API and MySQL backend
- Designed responsive UI with standard CSS

AIBG4-AlienWarz-Bot – AI Game Bot

Team Project – AI BattleGround Hackathon v4.0

GitHub: github.com/MihajloMilojevic/AIBG4-AlienWarz-Bot

Technologies: Python

- Developed an AI bot to play Alien Warz during a 24-hour hackathon
- Focused on pathfinding and strategic game-state evaluation

PQSAPwKR – Post-Quantum Stealth Address Protocol with Key Recovery

Cryptography Course Final Project

GitHub: github.com/MihajloMilojevic/PQSAPwKR/

Technologies: Python, Kyber, Post-Quantum Cryptography

- Researched and developed a stealth address protocol using Kyber
- Implemented a key recovery system compatible with post-quantum security assumptions
- Final project for advanced cryptography coursework

Competitions & Achievements

- 1st Place, EU Mobile App Competition Winter 2022
- 3rd Place, DigiHack Hackathon September 2024
- Finalist, AI BattleGround Hackathon v4.0 May 2024
- Finalist, National High School Hackathon December 2021
- 1st Place, National Electronics Competition May 2022
- Participant, math and programming competitions throughout high school (2019–2023)

Certificates

- Cryptography: Zero-Knowledge Proofs February 2025
- Cisco Networking Essentials April 2023
- Cisco IT Essentials: PC Hardware and Software June 2022
- Oracle Database Foundations May 2022
- Certifications – [freeCodeCamp Profile](#)
- Certifications – [HackerRank Profile](#)